Enhancement of scale factor accuracy of modulation algorithm for a closed-loop fiber-optic gyroscope.

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Abstract

TODO abstract

1 Motivation

iFOG principle of operation is based on the Sagnac effect. Its design has already been excercised for many years. However, as technology evolves, the limits holding the development back are pushed forward, enabling for better precision, miniaturized size or lower power consumption. At the same time, the ever progressing industry finds new and more strict applications for the gyroscopes, setting the requirements bar even higher. Such context makes the already existing designs being tested and optimized in every possible aspect. This applies equally to all the optical and electro–optic componenents, the electronics and the driving software. Crucial part of the control system of such gyroscope is the modulator. In a closed-loop design it is a vital element of the whole feedback loop, which directly affects the overall performance. It drives the optical circuit into a predefined state, in which the circuit response can be demodulated and error signals determined. Modulator accuracy has critical impact on the resulting performance. This paper takes a closer look at the all–digital implementation of the modulator subsystem and particularily on the algorithm controlling it. Mathematical analysis is conducted to allow for pure software enhancements in the algorithm accuracy. Then, by placing the algorithm in the context of various hardware configurations, its accuracy is also determined for each of them.

2 Principle of operation - open–loop configuration

A fibre–optic gyroscope utilizes Sagnac effect. It generates a light beam, which is ideally split in 50/50 ratio. Each half then enters a loop of relatively long fiber through each of its ends and propagates towards the other end. The fiber length L determines light transit time τ through it (Equation 1, where n is the refraction coefficient, affecting light speed in the medium).

$$\tau = \frac{L}{n \cdot c} \tag{1}$$

The halves go past each other and exit on the opposite ends of the fiber. The optical circuit is arranged, so that the two exitting beams interfere with each other and the resulting light intensity is measured at a detector. When the circuit stays at rest, the beams interfere with zero phase shift ($\Delta\phi$), resulting in the maximum intensity P_0 measured at the detector. However, when the circuit rotates, the two halves travel paths of slightly different lengths, resulting in a non–zero phase shift. The halves relative phase shift at the interference point is rotation dependent and in this paper it is reffered to as the Sagnac–induced phase shift:

$$\Delta\phi_s = \frac{2\pi \cdot L \cdot D \cdot n}{\lambda \cdot c} \cdot \omega \tag{2}$$

where D - fiber loop diameter, ω - angular velocity, λ - wavelength. Figure 1 depicts this phenomenon. Both halves enter the loop at the entry point. After the transit time they exit through the same point. However, because of non–zero rotation, this Entry/Exit point moves, effectively shrinking the path of one half and extending the path of the other. Path difference is relatively small, thus it is expressed as phase difference relative to the wavelength, rather than absolute length in meters.

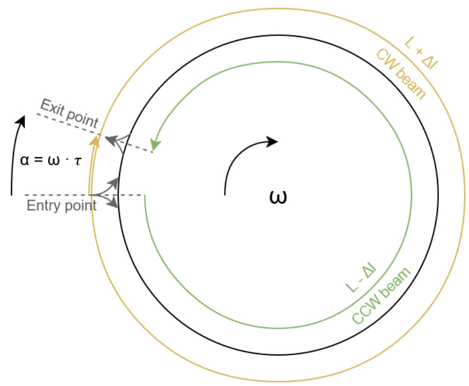


Figure 1: Two counterpropagating light beams enter the fiber loop at the *Entry point*. Their phase is initially in sync. After the transit time τ they exit and interfere. However, during that time, the *Entry/Exit point* moves according to rotation rate ω (by the angle $\alpha = \omega \cdot \tau$) and the CW and CCW beams covered non–equal path lengths ($L \pm \Delta l$). As a result they differ in phase.

The formula in Equation 3 and Figure 2 is the optical circuit response. It describes how the measured light intensity depends on the total phase shift.

$$P(\Delta\phi) = \frac{P_0}{2} \cdot (1 + cos(\Delta\phi)) \tag{3}$$

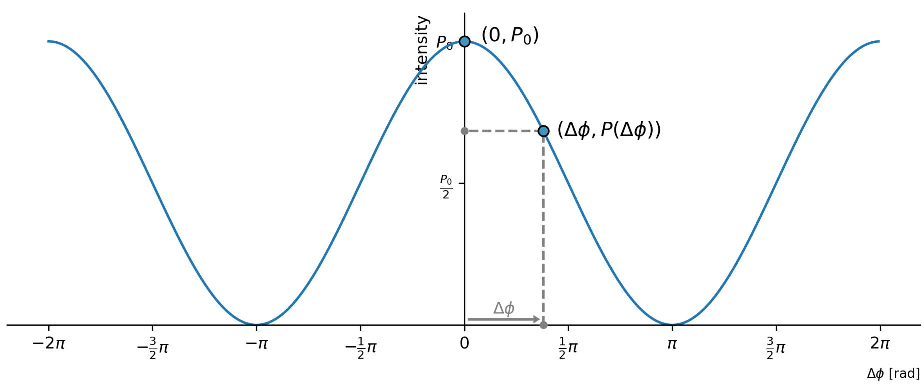


Figure 2: Optical circuit response function is a shifted cosine. The two exemplary operating points refer to the circuit being at rest, yielding the maximum possible intensity - in $(0, P_0)$ - and subjected to rotation, at non–zero phase shift and decreased intensity - in $(\Delta\phi, P(\Delta\phi))$.

When the circuit is at rest ($\omega = 0 \rightarrow \Delta\phi = 0$), the operating point resides at peak intensity P_0. A non–zero rotation results in a phase shift which moves the operating point according to rotation rate and direction, decreasing the measured intensity. Two problems are linked with such open–loop operation. Firstly, the intensity drop does not allow to tell the direction of rotation, as the response function is symmetrical. Secondly, in the resting region, the slope is flat, thus the sensitivity is zero. With greater rotation rate, the slope becomes steep, reaching its maximum sensitivity in $\pm\frac{\pi}{2}$, but then decreases again. For these reasons a closed–loop configuration will be superior in performance.

3 Modulation scheme - closed–loop configuration

In closed–loop configuration of an iFOG, the control system biases optical circuit sequentially in four predefined points - $\Delta\phi \in \{+\frac{3}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi, +\frac{1}{2}\pi\}$ - utilizing the *four–step modulation* (Figure 3). This allows for linearizing the response, keeping the circuit at maximum sensitivity and detecting the rotation direction.

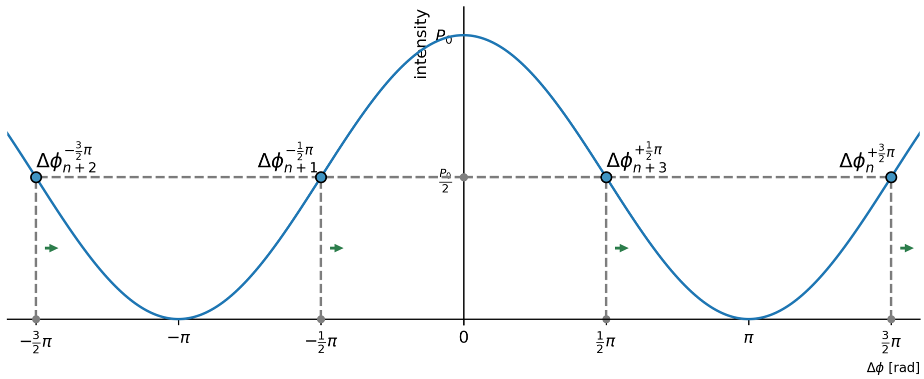


Figure 3: Circuit biased in four points of highest sensitivity.

Compared to *two–step modulation* this scheme is supreme, as it allows for controlling the scale factor. This is crucial due to several reasons. Modulator alters the phase of passing light through an electro–optic device, placed within the fiber loop. The applied phase shift is controlled by an electronic circuit. It performs digital calculations and applies a corresponding voltage using a digital–to–ananlog converter and an electronic front–end. Then the voltage is translated to a phase shift value by the electro–optic actuator. All of these elements are subjects temperature drift and aging, which affect the total combined gain. For this reason, constant calibration is required for maintaining measurement accuracy. In short, the control algorithm needs to know at all times where - voltage wise - are the desired bias points. Diagrams in Figure 4 depict how the biased circuit responds to various conditions - scale factor error and rotation present. From there, the control loop can determine how to adjust the parameters, to place the points back to where they shall ideally reside.

From the point of view of the control algorithm, the scale factor can be reffered to as the half–wave voltage V_π, even though it is not the only parameter affecting the effective scale factor. However, because of the assumed linearity of the other compounds, it is convenient to aggregate all the linear error sources into one parameter (other, non–linear errors, such as DAC INL are out of scope of this analysis).

Device control system biases the circuit continously with the following bias point order:

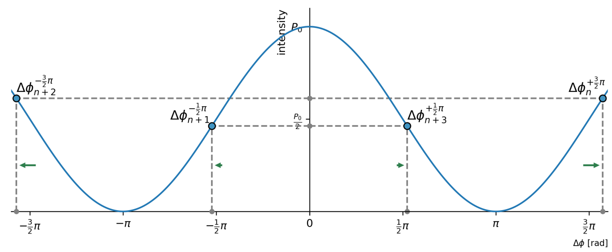
$$\{\Delta\phi_{4n}^{+\frac{3}{2}\pi}, \Delta\phi_{4n+1}^{-\frac{1}{2}\pi}, \Delta\phi_{4n+2}^{-\frac{3}{2}\pi}, \Delta\phi_{4n+3}^{+\frac{1}{2}\pi}\} \tag{4}$$

where the superscripted phase is the intended effective phase shift for the given sample.

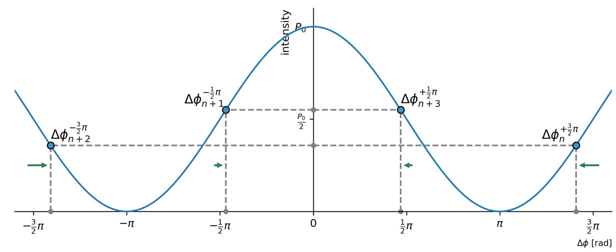
The modulator runs with the period of half the transit time τ. This allows for the effective phase shift to be a result of a relative shift between every k'th and $(k-2)$'th samples. For this reason the phase shift applied by the modulator for each sample can be half of the desired amplitude:

$$\begin{aligned}
\phi_{4n+0}^{+\frac{3}{2}\pi} &= +\frac{3}{4}\pi \cdot (1 + \delta_{4n+0}) \\
\phi_{4n+1}^{-\frac{1}{2}\pi} &= -\frac{1}{4}\pi \cdot (1 + \delta_{4n+1}) \\
\phi_{4n+2}^{-\frac{3}{2}\pi} &= -\frac{3}{4}\pi \cdot (1 + \delta_{4n+2}) \\
\phi_{4n+3}^{+\frac{1}{2}\pi} &= +\frac{1}{4}\pi \cdot (1 + \delta_{4n+3})
\end{aligned} \tag{5}$$

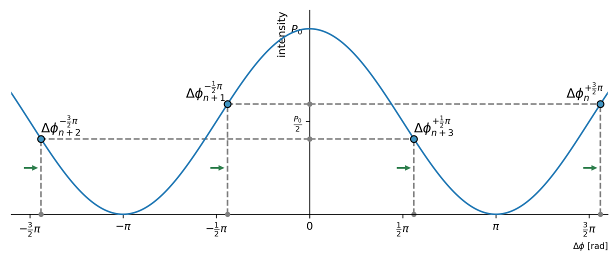
where δ_n denotes a close to zero scale factor error for the corresponding sample.



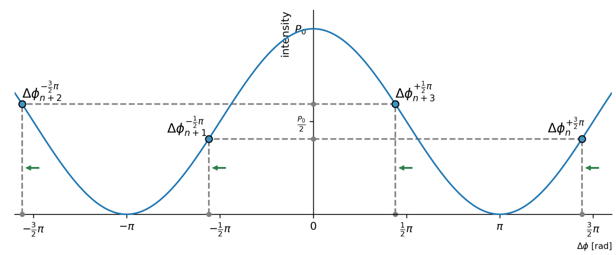
(a) Scale factor too high - bias points are shifted away from zero, resulting in intensity imbalance.



(b) Scale factor too low - bias points shifted towards zero, resulting in opposite imbalance.



(c) Phase shift induced by positive rotation - bias points shifted right. Points $\{4n, 4n+1\}$ correspond to higher intensity than points $\{4n+2, 4n+3\}$.



(d) Phase shift induced by negative rotation - bias points shifted left. Points $\{4n, 4n+1\}$ correspond to lower intensity than points $\{4n+2, 4n+3\}$.

Figure 4: Four–step modulation allows for detection of scale factor error and Sagnac–induced phase shift.

Additionally, to null the effect of the rotation–induced phase shift, the modulator adds an ever–updating ramp signal, utilizing Serrodyne modulation, with the update rate of $\Delta\phi_r = -\Delta\phi_s$ - equal in amplitude and opposite in sign. This way the algorithm cancells out the Sagnac effect, while knowing how much of it has been cancelled and thus it knows the exact rotation rate. In every cycle, the software monitors error inputs and adjusts control signals accordingly. A detailed operation of the control loop is described in TODO-lefevre?.

4 Modulator hardware architecture

This paper opts in for the usage of an all–digital modulator implementation. The architecture of this subsystem can be designed in several ways. Firstly - a fully analog variant, while feasible, does not comply with todays standards, where information is usually processed in digital form. For integration with external systems, an analog–to–digital interface would need to be present anyway. In such case, incorporating an MCU or a DSP in the processing loop brings the advantage of performing certain operations on pure data, rather than on signals carrying these data. In such context, a typical solution shall consist of a digital part for performing mathematical operations and several electronic elements for interfacing with optical circuit - this is the mixed digital/analog setup with a dedicated gain stage. From there an architecture with the minimal amount of electronic components can also be derived. It allows for implementing a software–defined modulator. The two latter are described in this section. The all–analog architecture will not be discussed.

4.1 Dedicated gain stage

A reasonable tradeoff between analog and digital approaches is the architecture proposed in TODO-lefevre?. It is composed of a digital part, a DAC and a dedicated gain stage. The configuration has been depicted in Figure 5.

In this configuration DAC and PGA are driven with an n–bit and m–bit words. The digital nature of controlling the gain stage introduces quantization noise into the scale factor control loop. One of the important advantages of this setup is the intrinsic separation of the modulation problem into two smaller

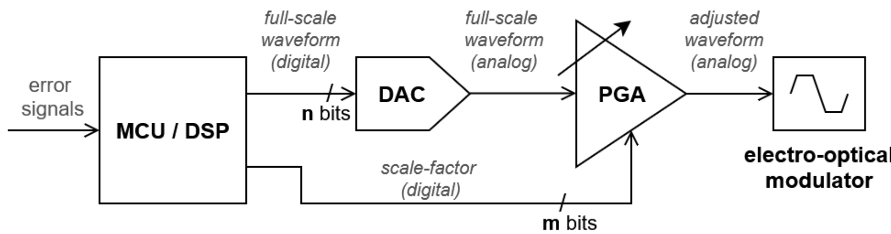


Figure 5: Modulator with a dedicated hardware gain stage. MCU/DSP outputs a fixed–scale digital waveform, which directly drives a DAC. Then, before reaching the electro–optic actuator, the analog waveform is multiplied by the scale factor by a programmable gain amplifier, or a custom gain stage, yielding an adjusted–scale waveform.

parts - generating the waveform in fixed–scale and applying the scale factor. The two can be implemented by software without dependence on each other. On the other side, the dedicated gain stage is a subject to interferences. It may have its own nonlinearities and introduce additional delay in the signal. Most importantly, its bandwidth may be gain dependent.

Utilizing the observation, that the scale factor shall not vary by more than a few per cent during the device's lifetime, it can be concluded that if the PGA was driven by $n+1$ bits, the performance would not be affected. Voltage range would have to be set for the worst–case scenario plus some margin, to always be sufficient. Full scale wouldn't be used most of the time. Even if the scale factor was to drop to 50% of the available range, which is beyond any pessimistic scenario, the overall performance would remain no worse than when using a n–bit control. From this an all–digital architecture can be defined.

4.2 All-digital approach - software–defined solution

The all–digital approach depicted in Figure 6 utilizes the availability and wide range of high precision digital–to–analog converters. It allows for dropping the physical gain stage, by applying scale factor digitally, prior to driving the DAC. This requires DAC reference voltage to be high enough to cover all possible ranges during the device lifetime. It also introduces quantization noise. However, both these disadvantages, are also problems of the mixed approach. The main advantage this idea brings is priparily the minimal amount of electronic components. This yields less sources of potential nonlinearities and electromagnetic interference. Since the electro–optic modulator has high input impedance, the output of a voltage–DAC can be directly connected to it, omitting any voltage buffers on the signal path, further reducing noise sources. While it may be safely assumed, that the profit resulting from less noisy electronics copensate DAC resolution loss, the following part of this paper discusses an algorithmic, which regains one bit of accuracy, when using the all–digital modulator architecture.

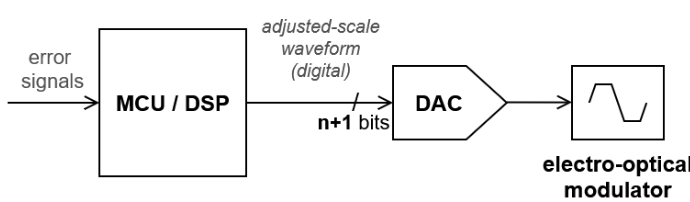


Figure 6: .

Additional advantage of leaning towards the proposed architecture is the fact, that the whole design becomes a software–defined solution. As in other fields, it brings flexibility from the R&D process perspective. While iterating hardware design is a time–consuming process, updating software is relatively quick. Offloating the whole problem to the software side, obviously makes its implementation more complex, but it limits the amount of interaction at the boundary of software and hardware developers. From this perspective the net workload remains at the same order of magnitude.

4.2.1 Voltage reference

For an all–digital modulator an adequate reference voltage has to be set up to account for the maximum possible scale factor. From Equation 5 it results, that a span of 1.5π is required for biasing and additionally, the Serrodyne modulator requires a span of 2π. This yields the required span to be $S = 3.5\pi$. Minimum DAC reference V_{ref}^{min} shall account for that with some margin $M \in [1.0; 1.1]$.

$$V_{ref}^{min} = V_{\pi}^{max} \cdot S \cdot M \tag{6}$$

The $V\pi^{max}$ denotes the highest expected half–wave voltage value. When determining final reference voltage it is advantageous to round it up to a multiple of one of typically available chip references (Equation 7), e.g. $N \cdot 5V$ or $N \cdot 4.096V$. It makes implementing the elctronic circuit more straightfroward. In practice, a fair compromise is to utilize the resulting voltage excess for the margin M, so that after adding it, the $V_{ref} \approx V_{ref}^{min}$.

$$V_{ref} = \lceil V_{ref}^{min} \rceil \tag{7}$$

4.2.2 Effect of the quantization error

Usage of a DAC results in quantization error. Converter resolution and its reference voltage determine the minimal controllable voltage change:

$$\Delta v = V_{ref} \cdot 2^{-N} \tag{8}$$

Relative error of the scale factor results from the distance between the controlled value and the actual. It corresponds to half of Δv. It can vary in time, as the actual V_π may continously change over time, but rather slowly (compared to the possible modulation frequencies).

$$\delta_q = \frac{1}{2} \cdot \frac{\Delta v}{V_\pi} \tag{9}$$

Finally, the scale factor error due to quantization depends on a few parameters according to the following formula:

$$\delta_q = \frac{1}{2} \cdot \frac{\lceil V_{\pi}^{max} \cdot S \cdot M \rceil \cdot 2^{-N}}{V_\pi} \tag{10}$$

Exemplary calculation, based on real–world values may yield:

$$\begin{aligned} \delta_q^{ex} &= \frac{1}{2} \cdot \frac{\lceil \overbrace{4.5V}^{V_\pi^{max}} \cdot \overbrace{3.5}^{S} \cdot \overbrace{1.025}^{M} \rceil \cdot \overbrace{2^{-16}}^{2^{-N}}}{\underbrace{3.5V}_{V_\pi}} = \frac{\lceil 16.14V \rceil \cdot 2^{-16}}{7V} = \frac{\overbrace{16.384V}^{4 \cdot 4.096V} \cdot 2^{-16}}{7V} = \\ &= 2^{-16} \cdot 2.34 = 2^{-16} \cdot 2^{1.23} \end{aligned} \tag{11}$$

The scale factor error can be compensated by the number of bits of the converter N, but the resolution loss will always stay at the level of over one bit. It must be noted, that most of this loss is due to the span S. It is required for proper operation of the algorithm and is not a consequence of the proposed all–digital architecture.

5 ??

To close the feedback loop, scale factor error signal is required. It is calculated from the incident light intensity difference between two consecutive bias points in each pair. The phase shift difference $\Delta\phi_k$, affecting intensity for the sample k, results from the current point drive ϕ_k and the drive two points prior.

$$\Delta\phi_k = (\phi_k - \phi_{k-2}) + \underbrace{\Delta\phi_s(k) + \Delta\phi_r(k)}_{=0} \tag{12}$$

The values of $\Delta\phi_s(n)$ and $\Delta\phi_r(n)$ denote the Sagnac–induced phase shift and the cancelling ramp shift, which are assumed to be zero for this analysis. By using equations 3 and 5 and assuming the *cos* function linear near the bias points (Equation 14), we get intensity values at each bias point:

$$\begin{aligned}
P_{4n+0}^{+\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + cos(\Delta\phi_{4n+0}^{+\frac{3}{2}\pi})) = \frac{P_0}{2} \cdot (1 + cos(\phi_{4n+0}^{+\frac{3}{2}\pi} - \phi_{4n-2}^{-\frac{3}{2}\pi})) = \\
&= \frac{P_0}{2} \cdot (1 + cos((\frac{3}{4}\pi \cdot (1 + \delta_{4n+0})) - (-\frac{3}{4}\pi \cdot (1 + \delta_{4n-2})))) = \\
&= \frac{P_0}{2} \cdot (1 + cos(\frac{3}{2}\pi \cdot (1 + \frac{\delta_{4n+0} + \delta_{4n-2}}{2}))) = \\
&= \frac{P_0}{2} \cdot (1 + \frac{3}{2}\pi \cdot \frac{\delta_{4n+0} + \delta_{4n-2}}{2}) \\
P_{4n+1}^{-\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 - \frac{1}{2}\pi \cdot \frac{\delta_{4n+1} + \delta_{4n-1}}{2}) \\
P_{4n+2}^{-\frac{3}{2}\pi} &= \frac{P_0}{2} \cdot (1 + \frac{3}{2}\pi \cdot \frac{\delta_{4n+2} + \delta_{4n+0}}{2}) \\
P_{4n+3}^{+\frac{1}{2}\pi} &= \frac{P_0}{2} \cdot (1 - \frac{1}{2}\pi \cdot \frac{\delta_{4n+3} + \delta_{4n+1}}{2})
\end{aligned} \tag{13}$$

$$cos(\phi_B + \Delta) = \underbrace{cos(\phi_B)}_{=0} + cos'(\phi_B) \cdot \Delta = -sin(\phi_B) \cdot \Delta \qquad \text{for } \phi_B \in \{+\frac{3}{2}\pi, +\frac{1}{2}\pi, -\frac{1}{2}\pi, -\frac{3}{2}\pi\} \tag{14}$$

Intensity difference for the samples $4n + 1$ and $4n + 3$ is:

$$\begin{aligned}
e_{4n+1}^{-\frac{1}{2}\pi} &= P_{4n+1}^{-\frac{1}{2}\pi} - P_{4n+0}^{+\frac{3}{2}\pi} = -\frac{P_0 \cdot \pi}{8} \cdot (\delta_{4n+1} + 3 \cdot \delta_{4n+0} + \delta_{4n-1} + 3 \cdot \delta_{4n-2}) \\
e_{4n+3}^{+\frac{1}{2}\pi} &= P_{4n+3}^{+\frac{1}{2}\pi} - P_{4n+2}^{-\frac{3}{2}\pi} = -\frac{P_0 \cdot \pi}{8} \cdot (\delta_{4n+3} - 3 \cdot \delta_{4n+2} + \delta_{4n+1} + 3 \cdot \delta_{4n+0})
\end{aligned} \tag{15}$$

which can be generalized for every odd bias point (scale factor error shall only be calculated at odd points - this results from the demodulation algorithm assumptions):

$$e_{2k+1} = -\frac{P_0 \cdot \pi}{8} \cdot (\underbrace{\delta_{2k+1}}_{\text{current error}} + \underbrace{3 \cdot \delta_{2k+0} + \delta_{2k-1} + 3 \cdot \delta_{2k-2}}_{\text{last three errors}}) \tag{16}$$

From this it can be seen, that the error signal, used for control loop closure, is dependent on four recent drive values.

References

[Doe] *First and last LaTeX example.*, John Doe 50 B.C.

ϕ_{2k+1} ($\times 1$)	ϕ_{2k+0} ($\times 3$)	ϕ_{2k-1} ($\times 1$)	ϕ_{2k-2} ($\times 3$)	$e_{2k+1}[-\frac{P_0 \cdot \pi}{8} \cdot \Delta\phi_{min}]$
0	0	0	0	0
$\Delta\phi_{min}$	0	0	0	1
0	0	$\Delta\phi_{min}$	0	
$\Delta\phi_{min}$	0	$\Delta\phi_{min}$	0	2
0	$\Delta\phi_{min}$	0	0	3
0	0	0	$\Delta\phi_{min}$	
$\Delta\phi_{min}$	$\Delta\phi_{min}$	0	0	4
0	$\Delta\phi_{min}$	$\Delta\phi_{min}$	0	
0	0	$\Delta\phi_{min}$	$\Delta\phi_{min}$	
$\Delta\phi_{min}$	0	0	$\Delta\phi_{min}$	
$\Delta\phi_{min}$	$\Delta\phi_{min}$	$\Delta\phi_{min}$	0	5
$\Delta\phi_{min}$	0	$\Delta\phi_{min}$	$\Delta\phi_{min}$	
0	$\Delta\phi_{min}$	0	$\Delta\phi_{min}$	6
$\Delta\phi_{min}$	$\Delta\phi_{min}$	0	$\Delta\phi_{min}$	7
0	$\Delta\phi_{min}$	$\Delta\phi_{min}$	$\Delta\phi_{min}$	
$\Delta\phi_{min}$	$\Delta\phi_{min}$	$\Delta\phi_{min}$	$\Delta\phi_{min}$	8

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